

The fixed algebra on the CAR trace Hilbert space

`MathlibAnnex.CStarAlgebra.CAR.separableCounterexampleRepresentation`

def

Defines a representation of the same algebra T on the GNS Hilbert space of the CAR trace.

Statement

Let A be the completed CAR algebra and $T = C^*(\pi(A) \cup \{L_i : i \in I\}) \subseteq B(\mathcal{H})$ the algebra generated by the selected direct sum π of pure-state GNS representations indexed by their unitary-equivalence classes $i \in I$ and the unitaries L_i from the fixed homogeneity and shell construction. Write $\iota(a) = \pi(a)$ for the embedding $A \rightarrow T$. Let τ be the normalized CAR trace, $H_\tau = L^2(A, \tau)$ its GNS Hilbert space, and π_τ its GNS representation. Let ϕ be the chosen state on T with $\phi \circ \iota = \tau$, and let $(H_\phi, \rho_\phi, \xi_\phi)$ be its GNS representation with cyclic vector. The cited trace construction provides a chosen unitary $E : H_\tau \rightarrow H_\phi$ carrying the canonical trace vector to ξ_ϕ and intertwining the two representations of A . Define

$$\tilde{\rho}(t) = E^{-1} \rho_\phi(t) E \quad (t \in T).$$

Definition

The representation is obtained by conjugating ρ_ϕ with the same chosen unitary $E : H_\tau \rightarrow H_\phi$: $\tilde{\rho}(t) = E^{-1} \rho_\phi(t) E$. Its operators therefore act on H_τ , not on H_ϕ .

Assumptions

The algebra T and the state extension ϕ are fixed as in the cited construction. The same unitary E is used throughout. The Hilbert space H_τ is determined by A and τ , independently of the added unitaries L_i .

Conclusion

This is a unital $*$ -representation $\tilde{\rho} : T \rightarrow B(H_\tau)$. It extends the CAR trace representation in the sense that $\tilde{\rho}(\iota(a)) = \pi_\tau(a)$ for all $a \in A$.

The space H_τ is separable by its dense CAR orbit, and injectivity is a separately cited theorem. Neither fact asserts norm separability of T or irreducibility of this representation.

Main citations

- Definition and its exact construction —
`MathlibAnnex.CStarAlgebra.CAR.separableCounterexampleRepresentation`
- The CAR trace Hilbert-space abbreviation —
`MathlibAnnex.CStarAlgebra.CAR.SeparableCounterexampleHilbertSpace`

- The transported representation and its exact RHS —
`MathlibAnnex.CStarAlgebra.CAR.traceModelRepresentation`
- The pointwise conjugation formula —
`MathlibAnnex.CStarAlgebra.CAR.traceModelRepresentation_apply`
- Its source restriction —
`MathlibAnnex.CStarAlgebra.CAR.traceModelRepresentation_shellFamilySourceHom`
- Separability of the trace Hilbert space —
`MathlibAnnex.CStarAlgebra.CAR.separableSpace_traceHilbertSpace`

Supporting route explanation

Lean source signature (exact)

```
noncomputable def separableCounterexampleRepresentation :
  Representation AtomicCounterexampleAlgebra SeparableCounterexampleHilbertSpace :=
  traceModelRepresentation homogeneityShellFamily
```

In the source	Mathematical meaning
AtomicCounterexampleAlgebra	The unchanged concrete target T of the fixed CAR homogeneity shell construction, with source embedding $\iota : A \rightarrow T$.
SeparableCounterexampleHilbertSpace	The CAR trace GNS space $H_\tau = L^2(A, \tau)$, determined by CAR and its normalized trace, not the atomic sum $\mathcal{H}$.
Representation AtomicCounterexampleAlgebra SeparableCounterexampleHilbertSpace	The output is a unital star representation $\tilde{\rho} : T \rightarrow B(H_\tau)$ of this same target.
traceModelRepresentation homogeneityShellFamily	The full RHS specializes the trace-model action to the same fixed family. With its chosen $E : H_\tau \rightarrow H_\phi$, it is $\tilde{\rho}(t) = E^{-1} \rho_\phi(t) E$. The source restriction is $\tilde{\rho}(\iota(a)) = \pi_\tau(a)$; injectivity and separability are separately cited facts, and irreducibility is not asserted.

Further source notes: The names `traceGNSUnitary` (E) and `tracialRepresentation` (ρ_ϕ) occur in the linked defining construction, explaining the conjugation although they are absent from this short signature.

Earlier published Card and PDF

▼ Exact content identity

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Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`